Achievements

- Finished **Top 15** among **200+ UG/PG students** in **QRT QuantQuest**, **IIT Bombay** (sole PG team) *(2024)*
- Secured **2nd place** in **F.I.R.E.** (Flame Imaging Robot Extinguisher) technical event at **Tirutsava** *(2020)*
- Secured **2nd place** in **ArduiKnow Challenge**, a competitive microcontroller event at **Tirutsava** *(2020)*
- Secured **3rd place** in **Terrace Farming Robot** design contest among **23 IITs** at **Inter IIT Tech Meet** *(2019)*

Professional Experience

Software Developer (Backend) | *Paytm* *(Jan '23 - Dec '23)*

FSM & BINARY Teams, Paytm, India

- Architected hierarchical, location-based **feature-flagging** backend, accelerating product feedback loops by **35-40%**
- Developed dynamic **territory access controls**, incentivizing agent productivity to drive a **19% increase** in resolutions
- Engineered **API orchestration** to cross-reference KYC databases, reducing vendor onboarding friction by **32%**
- Designed load-balanced **round-robin** automated ticketing algorithm, saving team leads **15+ hours/week** of overhead
- Automated manual finance workflows by **90%** using efficient **1-SAP logic** to process **15M revenue records** monthly

Jr. Data Scientist | *Holmusk* *(Jun '22 - Aug '22)*

- Architected predictive patient outcome models within the **NeuroBlu** webapp using **XGBoost and LSTM** architectures to model structured clinical and longitudinal patient data, improving prediction accuracy for healthcare analytics
- Optimized dashboard backend using **Python**, reducing query latency from **20s to 2s** (**10x speedup**)

Internship Experience

AI Research Intern | *Motilal Oswal Financial Services* *(May '26 - Aug '26)*

Align Before You Quantize: Time Series Tokenization

- Engineered **HALO** framework for **Leech lattice** time-series tokenization via geometric alignment for **AAAI**
- Reduced reconstruction MSE by **11.9%**(IN) & **2.8%**(US) via phased training, boosting codebook usage **3.89x**
- Enhanced downstream forecasting via discrete representations, achieving SOTA **IC** & cutting **MASE** by **7.5%**

Key Projects

Implied Volatility Surface Forecasting *(Aug '25 - Aug '26)*

Master's Thesis | IITB Citadel Research Lab Project |(Submitted to 7th Int. Conf. on AI in Finance)

- Spearheaded research on no-arbitrage implied volatility surface forecasting using GANs and Diffusion Models, evaluating **AF-TransVol** and **AF-INO** architectures across **3,417 trading days** of NIFTY50 data
- Engineered **44 causal scalars** & **3 soft regimes** using scale-invariant log-IV transforms & (ρ, z) warping
- Enforced butterfly convexity & calendar monotonicity, achieving SOTA **5.30% MAPE** and $R^2 = \mathbf{0.910}$
- Slashed static arbitrage violations by **>50%** using stress-conditioned **AF-TransVol-Stress** extensions

Tick-Ops: Quant Data Engine Evaluation | *Self Project* | *DuckDB, QuestDB* *(Jan '26 - May '26)*

- Benchmarked **DuckDB, QuestDB, ClickHouse** against **Pandas** across 5 tick workloads with **1.05M quotes**
- Built deterministic verification oracle enforcing strict float tolerances (**1e-12**) to ensure exact accuracy
- Demonstrated **19.5x compute speedups** via **DuckDB**, leveraging sorted row-groups to skip data partitions
- Validated order book state integrity, resolving boundary leaks using monotonic sequence IDs & timestamping

Lock-Free MPMC Queue | *Self Project* | *Low-Latency C++20* *(Aug '25 - Nov '25)*

- Architected high-throughput **MPMC queue** for UDP feeds, strictly isolating network I/O from payload parsing
- Implemented **C++20 lock-free ring buffer** utilizing **CAS atomics** & memory barriers to block race conditions
- Eliminated OS-level thread blocking via per-slot sequence numbers, ensuring highly concurrent memory access
- Profiled latency via **Python**, slashing median wait times by **56%** and p99 tail latency to **25µs**

Technical Skills

Languages	Python, C/C++, JavaScript, Java, Bash, SQL, Markdown
ML & Data Science	PyTorch, TensorFlow, JAX, Scikit-Learn, Keras, OpenCV, NumPy, Pandas, SciPy
Web & Utilities	Django, Spring Boot, React.js, Flask, Matplotlib, Seaborn, BeautifulSoup, Selenium
Tools & Platforms	Git, Docker, AWS, Linux, MATLAB, LaTeX, Pygame, p5.js, InkScape, Procreate